Random category-specific Tune-1 PnP for PCM

Matched test set 45, 20% Hadamard sampling, NPL-default noise

Completed reconstruction experiment, version 1.0, 9 November 2026

Ten requested methods; five random calibration choices per PnP family

1 Executive summary

We evaluate practical category-specific Tune-1 parameters selected *without ground truth* from noisy validation data. For each of PnP-ADMM and GS-PnP, we draw one of five frozen calibration configurations *independently* for each device category. These choices are frozen before test metrics are examined. The same selected configuration is applied to all nine test images from that category. Test set 45 is a fixed stratified subset (9 images per category) of the original test set 450 (90 images per category).

All ten requested methods use the *same frozen noisy measurement and Hadamard mask for each test image*, with 20% multilevel sampling, $J = 8$, coarse $J_c = 6$ and the original NPL-default PCM noise realisation. The two new methods contribute 45 reconstructions each, with no additional parameter search on test data.

Main result: On 45 matched test images, the randomly chosen PnP-ADMM Tune-1 configuration averaged 35.342 dB, compared with 35.554 dB for PnP-ADMM-tuned-5 (paired difference -0.212 dB). GS-PnP Tune-1 averaged 32.426 dB, compared with 32.336 dB for GS-PnP-tuned-5 (paired difference +0.089 dB).

We report PSNR, SSIM, HaarPSI and MSE on raw (unclipped) reconstructions. These are *retrospective* measurements: the full test set has been examined in related earlier studies, and one random draw alone cannot establish the variability of Tune-1 performance across calibration choices.

2 Random selection and experimental protocol

All five validation configurations per category were already selected separately for both PnP families using four-fold SSDU measurement-prediction loss and log-space ternary search. The 25 ADMM configurations and 25 GS configurations were *frozen before this new comparison.*

We made ten independent uniform draws, one per (category, PnP family), with deterministic random seed `PCM-test45-one-calibration-2026-11-09-v1`. The selection file stores the five eligible calibration IDs in sorted order, selected index, exact frozen parameter values and each source file's SHA256. No image PSNR or category-level reconstruction metric informed the random choice.

	ADMM Tune-1				GS-PnP Tune-1				
Category	Calibration ID	ρ	σ	SSDU risk	Calibration ID	λ	σ	τ_0	SSDU risk
CIGS	0768	0.78297	0.05196	0.0038248	0530	0.09283	0.06393	4.833	0.0036331
Organic	0768	0.78297	0.07862	0.0043211	0158	0.04850	0.04743	46.554	0.0042415
Perovskite	0008	0.78297	0.05196	0.0044673	0768	0.05705	0.05391	4.833	0.0037019
Reference	0008	0.93853	0.05196	0.0052309	0008	0.20899	0.06393	28.897	0.0051983
Silicone	0008	0.65320	0.12456	0.0052620	0008	0.09283	0.10665	22.767	0.0055016

Table 1: One calibration configuration randomly drawn from five validation-tuned choices in each category. Each SSDU risk is the held-out measurement-prediction error for that individual validation acquisition, not an image-space test MSE.

New reconstructions: PnP-ADMM on RTX 4070 and GS-PnP on RTX 4090, each with unchanged pretrained denoiser weights and original solver stopping rules. Model and measurement normalisation were not altered between tuning and testing. The test45 measurement and pattern SHA256s were checked against the historical full test set 450 archive.

3 Methods compared

We report exactly ten method labels: zero-filled, SPGL1-BP, SPGL1-BPDN, PnP-ADMM-tuned-5, GS-PnP-tuned-oracle, scalar TV, UTV-C60, GS-PnP-tuned-5, PnP-ADMM Tune-1 and GS-PnP Tune-1. The supervised-validation-selected *PnP-ADMM oracle label* is intentionally omitted. PnP-ADMM-tuned-5 is shown

once, even though the original SSDU-tuned-5 settings coincided with that reference. The GS-PnP expanded SSDU-5 method is also intentionally omitted.

The eight historical reference scores are taken from the same fixed 450-image benchmark after checking each category, image ID, noisy measurement hash and sampling mask hash. Their established hyperparameters are unchanged. Missing historical raw-image arrays were rerun solely for visualisation under those original numerical settings. SPGL1-BP can show small differences in a later numerical replay: the archived benchmark metrics remain authoritative for the main comparison, and visual captions for BP use metrics calculated on the displayed replayed array. No baseline parameter was adjusted to fit test results.

4 Quantitative results

Overall metrics, 45 unique test images

Method	PSNR (dB)	SSIM	HaarPSI	MSE
Zero-filled	24.97 ± 1.95	0.4822 ± 0.0506	0.5522 ± 0.0483	3.517e-03 ± 1.628e-03
SPGL1-BP	25.39 ± 1.47	0.4359 ± 0.0316	0.6464 ± 0.0705	3.065e-03 ± 1.112e-03
SPGL1-BPDN	29.28 ± 3.35	0.7126 ± 0.1040	0.6593 ± 0.0950	1.549e-03 ± 1.106e-03
PnP-ADMM tuned-5	35.55 ± 3.59	0.8957 ± 0.0803	0.8810 ± 0.0609	3.752e-04 ± 2.610e-04
GS-PnP oracle	34.83 ± 3.52	0.8656 ± 0.0869	0.8608 ± 0.0699	4.323e-04 ± 2.750e-04
Scalar TV	29.97 ± 1.92	0.7178 ± 0.0560	0.7824 ± 0.0509	1.110e-03 ± 5.161e-04
UTV-C60	32.15 ± 2.99	0.8391 ± 0.0950	0.7632 ± 0.1123	7.564e-04 ± 4.776e-04
GS-PnP tuned-5	32.34 ± 2.76	0.7476 ± 0.0977	0.8193 ± 0.0750	7.059e-04 ± 4.210e-04
PnP-ADMM Tune-1	35.34 ± 3.82	0.8904 ± 0.0891	0.8607 ± 0.0786	4.103e-04 ± 3.059e-04
GS-PnP Tune-1	32.43 ± 2.78	0.7358 ± 0.0989	0.8373 ± 0.0699	6.897e-04 ± 3.996e-04

Table 2: Means ± sample standard deviations over 45 *distinct* images. This does not count the ten random calibration draws as additional test images.

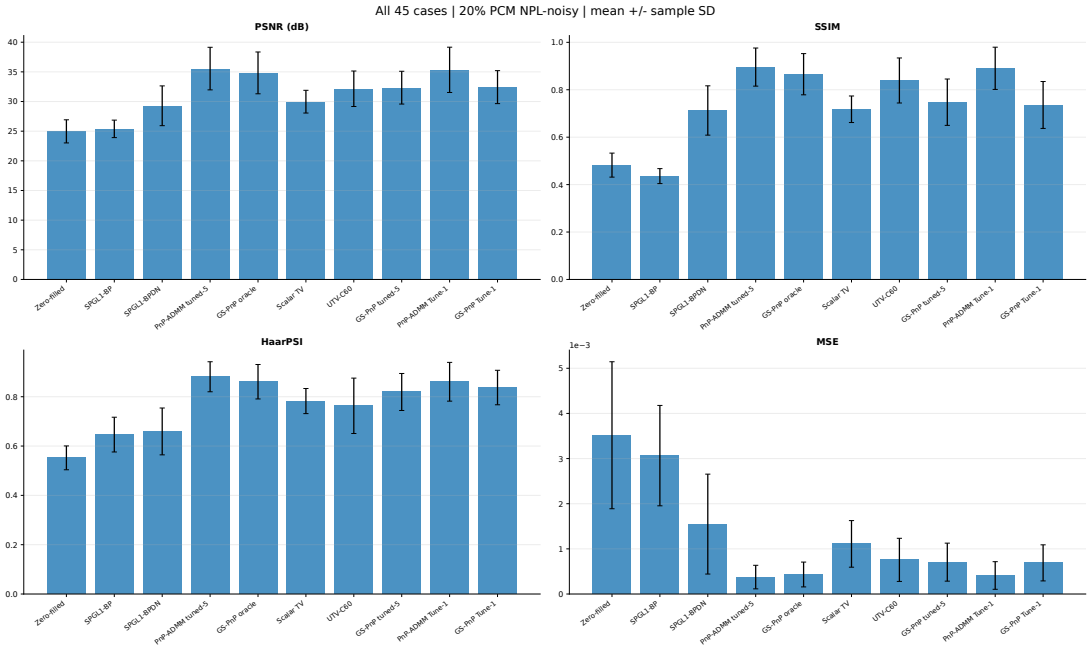


Figure 1: Overall PSNR, SSIM, HaarPSI and MSE means with sample standard-deviation whiskers.

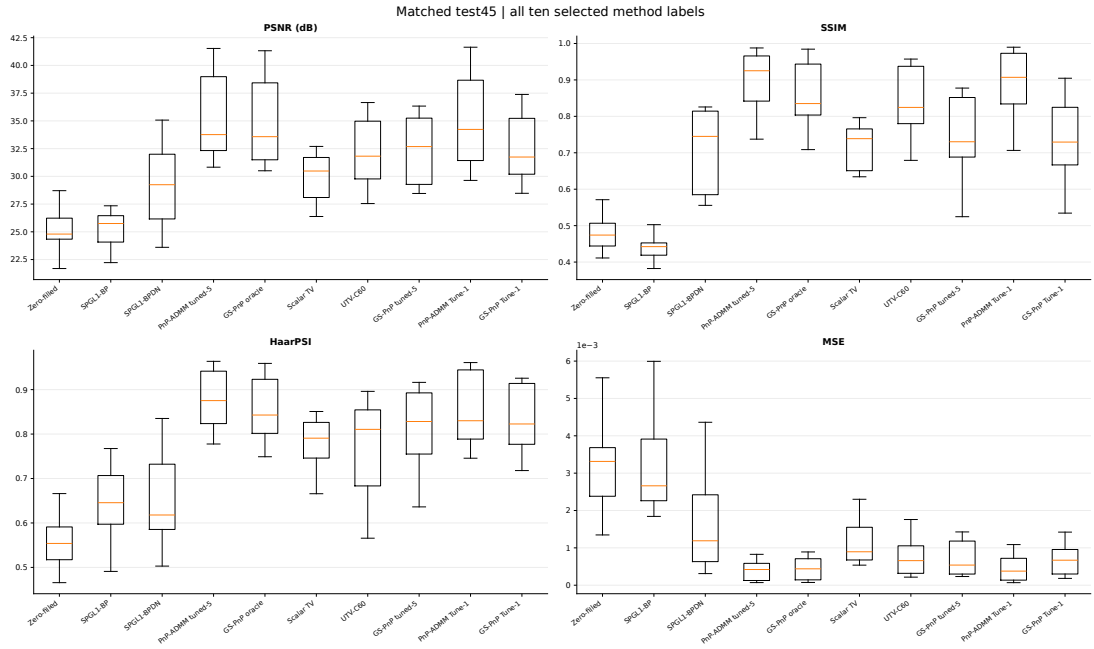


Figure 2: Distribution of 45 paired test-image scores; outliers are hidden in the plot, but used in all means.

Per-category PSNR

Category	ZF	SPGL1-BP	BPDN	ADMM-5	GS oracle	TV	C60	GS-5	ADMM Tune-1	GS Tune-1
CIGS	25.75	26.53	31.27	39.75	38.10	31.22	33.43	35.34	39.53	35.35
Organic	25.59	26.32	31.78	38.37	37.79	31.46	35.36	34.84	38.57	35.33
Perovskite	26.98	26.35	31.48	35.29	35.06	31.23	32.46	31.09	34.51	32.03
Reference	24.37	24.42	26.67	31.91	31.57	28.24	29.22	29.51	31.72	29.61
Silicone	22.15	23.33	25.21	32.46	31.62	27.72	30.29	30.90	32.38	29.80

Table 3: Mean PSNR in dB for nine fixed test images per category. Companion figures report SSIM, HaarPSI and MSE, as well as sample SD.

Tune-1 versus the previous tuned-5 baseline

	ADMM PSNR (dB)			GS-PnP PSNR (dB)		
Category	Tuned-5	Tune-1	Difference	Tuned-5	Tune-1	Difference
CIGS	39.75	39.53	-0.22	35.34	35.35	+0.00
Organic	38.37	38.57	+0.20	34.84	35.33	+0.49
Perovskite	35.29	34.51	-0.78	31.09	32.03	+0.95
Reference	31.91	31.72	-0.19	29.51	29.61	+0.11
Silicone	32.46	32.38	-0.07	30.90	29.80	-1.10

Table 4: Paired category mean PSNR differences, Tune-1 minus the existing SSDU tuned-5 configuration. Positive means Tune-1 obtained higher PSNR on average for that category's nine test images.

Reconstruction runtime

For the 45 newly reconstructed test images, the observed mean ± SD was 0.820 ± 0.448 seconds/image for ADMM Tune-1 and 1.162 ± 0.452 seconds/image for GS-PnP Tune-1. These exclude the earlier calibration-search cost and do not constitute a controlled speed comparison because the algorithms ran on different GPUs.

5 Visual comparisons and reproducibility

The companion 45-page PDF shows *every* image in test set 45, with ground truth and exactly the ten methods specified above. Each method panel displays PSNR, SSIM, HaarPSI and MSE. The display uses the AFM-hot colormap capped at 0.8, with the same image intensity range $[0, 1]$ for every panel.

Files retained: a frozen random-selection JSON; the original test45 image-ID and acquisition-hash manifests; 45 individual new ADMM reconstructions and 45 new GS-PnP reconstructions with metrics and solver diagnostics; archived benchmark scores and replayed visual arrays; a 45-row ten-method metrics CSV; per-category and global JSON summaries; bar/boxplot PDFs and five nine-page category visual PDFs. All experiment files are versioned separately from prior tune-1 calibration studies and test set 450.

6 Conclusion

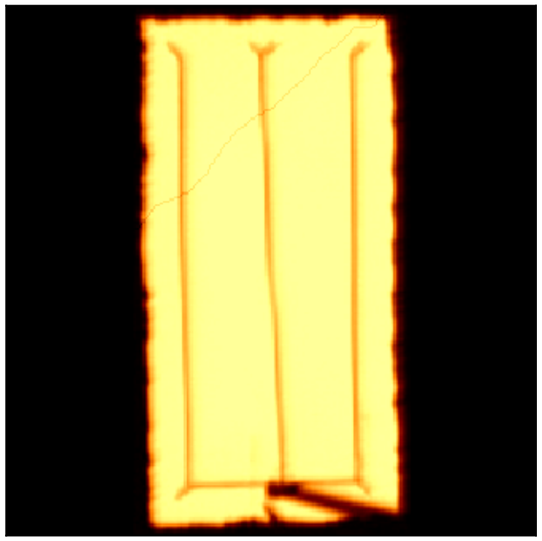
On 45 matched test images, the randomly chosen PnP-ADMM Tune-1 configuration averaged 35.342 dB, compared with 35.554 dB for PnP-ADMM-tuned-5 (paired difference -0.212 dB). GS-PnP Tune-1 averaged 32.426 dB, compared with 32.336 dB for GS-PnP-tuned-5 (paired difference +0.089 dB).

A single category-specific calibration can yield reasonable reconstruction quality, but our results represent *one uniform random choice per category and method*. They do not establish that one calibration image is enough in general, that the joint PnP parameter objective is globally unimodal, or that SSDU loss predicts the best image-space reconstruction. The test45 comparisons must not be used to repeatedly retune parameters.

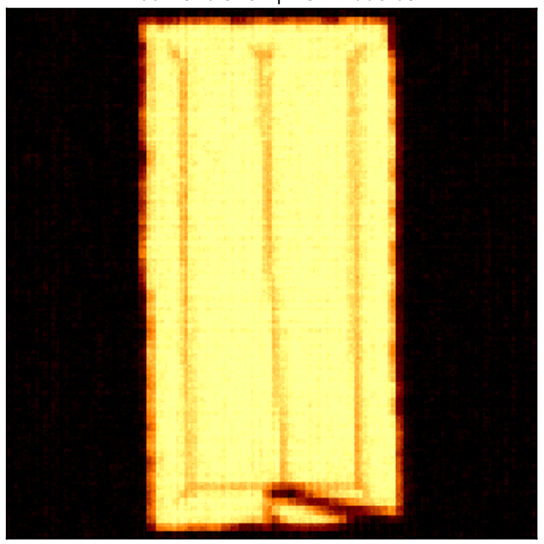
Primary references Yaman et al. (2020), acquired-measurement splitting for SSDU: https://doi.org/10.1002/mrm.28378. Hurault, Leclaire and Papadakis (2022), GS-PnP algorithm and backtracking: https://openreview.net/pdf?id=fPhKeld3Okz.

CIGS | test image 0068 | 20% Hadamard + NPL-default noise

Target
Ground truth



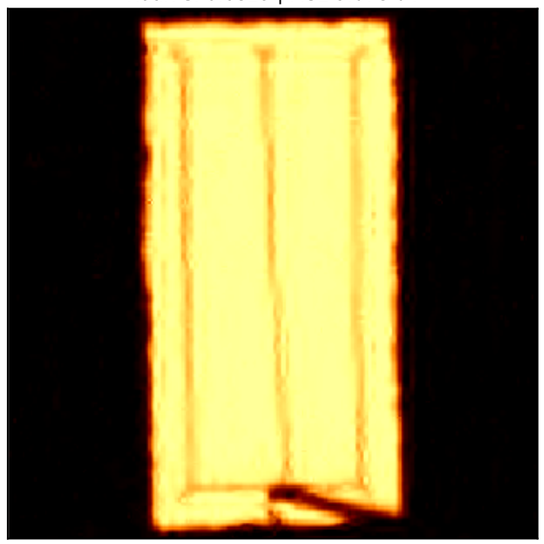
Zero-filled
PSNR 25.84 dB | SSIM 0.4814
HaarPSI 0.5787 | MSE 2.60e-03



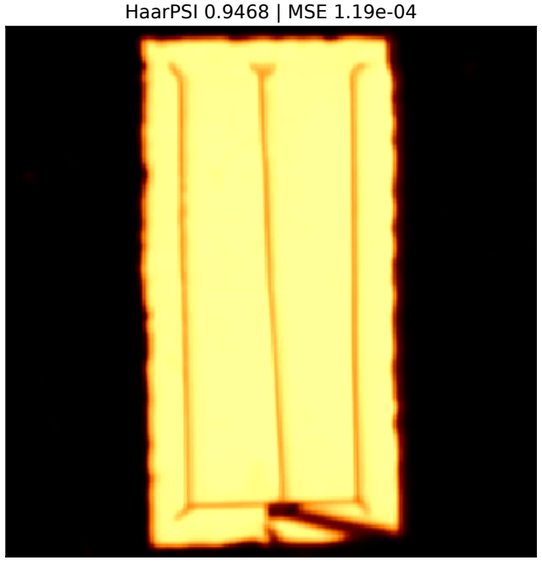
SPGL1-BP
PSNR 26.44 dB | SSIM 0.4456
HaarPSI 0.6652 | MSE 2.27e-03



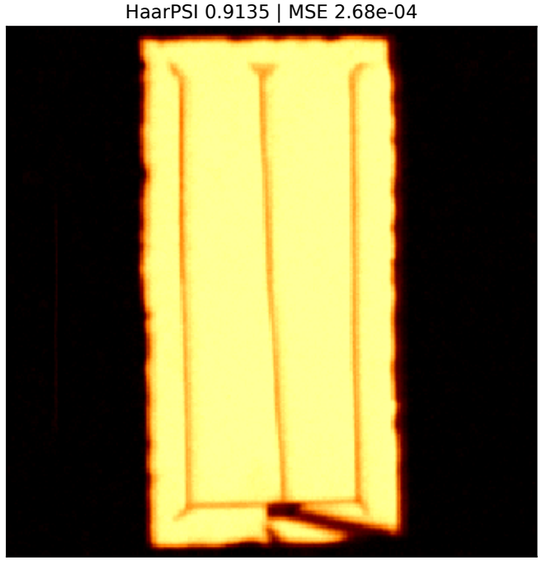
SPGL1-BPDN
PSNR 30.42 dB | SSIM 0.7745
HaarPSI 0.6829 | MSE 9.07e-04



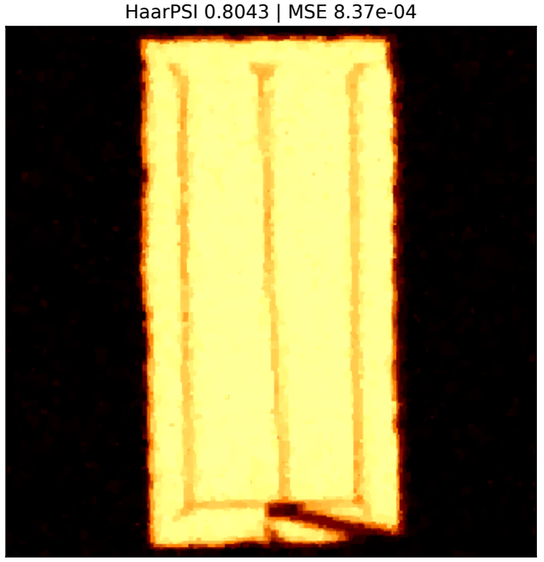
PnP-ADMM tuned-5
PSNR 39.23 dB | SSIM 0.9757
HaarPSI 0.9468 | MSE 1.19e-04



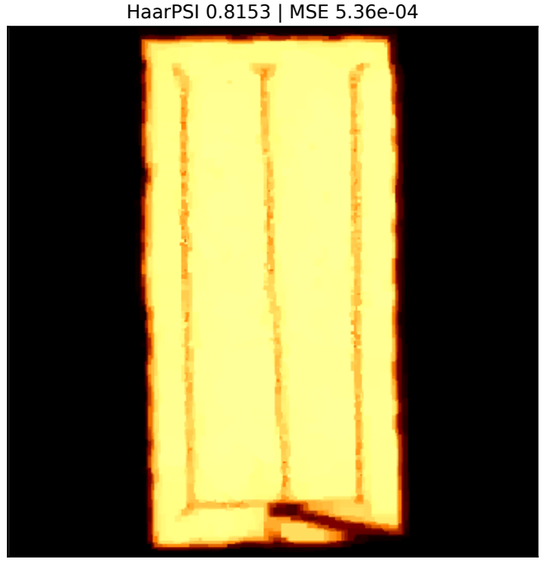
GS-PnP oracle
PSNR 35.72 dB | SSIM 0.8351
HaarPSI 0.9135 | MSE 2.68e-04



Scalar TV
PSNR 30.77 dB | SSIM 0.7427
HaarPSI 0.8043 | MSE 8.37e-04



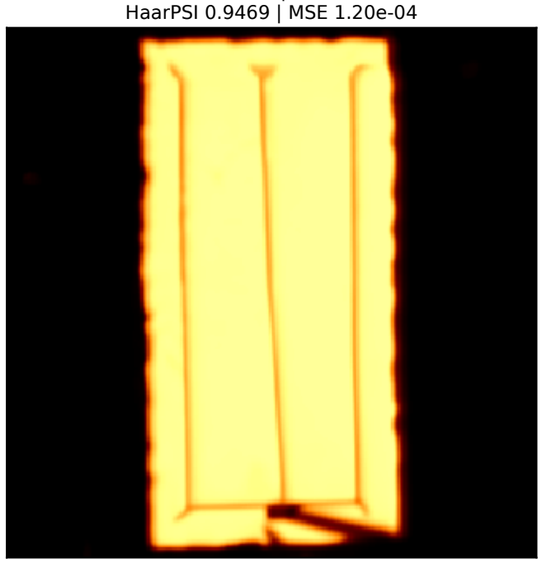
UTV-C60
PSNR 32.71 dB | SSIM 0.9347
HaarPSI 0.8153 | MSE 5.36e-04



GS-PnP tuned-5
PSNR 35.48 dB | SSIM 0.8618
HaarPSI 0.9044 | MSE 2.83e-04



PnP-ADMM Tune-1 (0768)
PSNR 39.22 dB | SSIM 0.9773
HaarPSI 0.9469 | MSE 1.20e-04

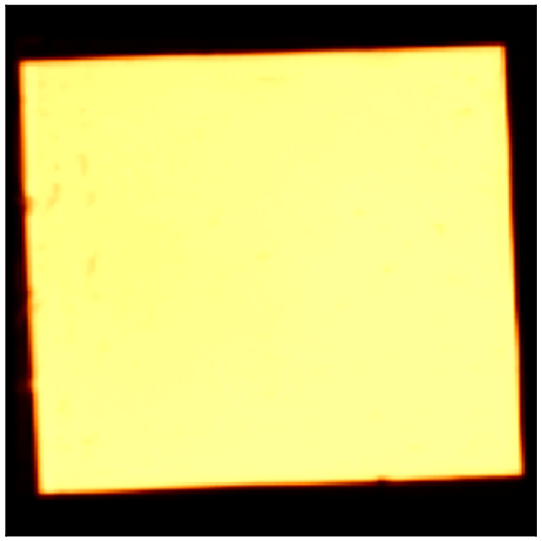


GS-PnP Tune-1 (0530)
PSNR 35.33 dB | SSIM 0.8225
HaarPSI 0.9122 | MSE 2.93e-04

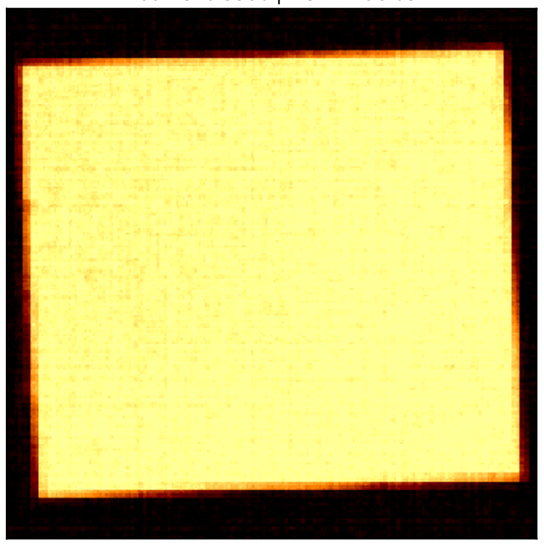


Organic | test image 0014 | 20% Hadamard + NPL-default noise

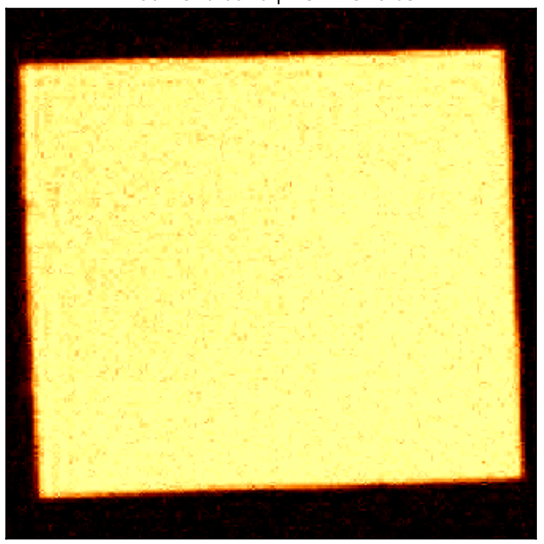
Target
Ground truth



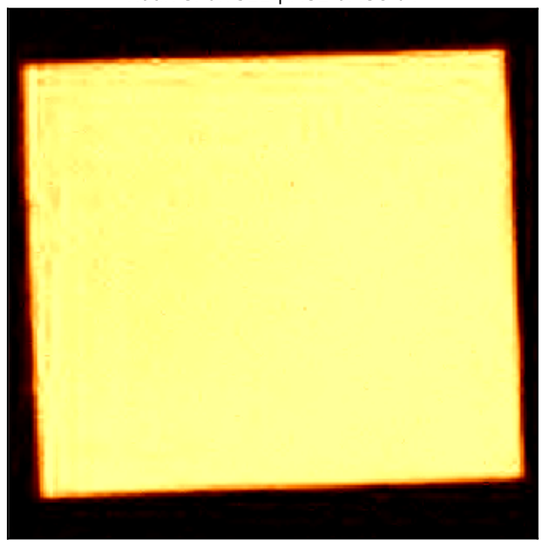
Zero-filled
PSNR 25.54 dB | SSIM 0.4741
HaarPSI 0.5586 | MSE 2.79e-03



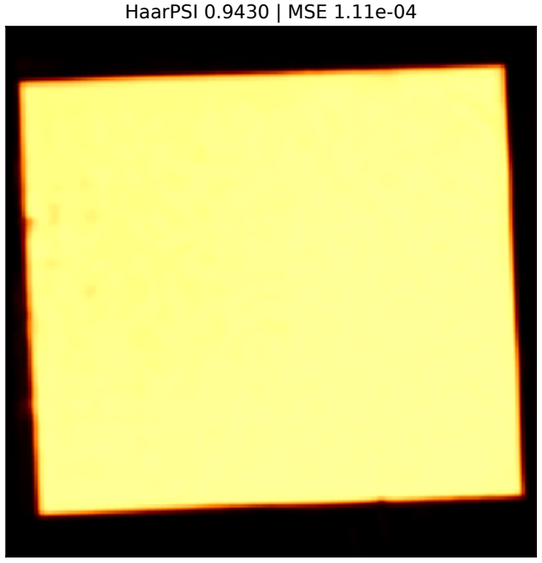
SPGL1-BP
PSNR 26.35 dB | SSIM 0.4204
HaarPSI 0.6940 | MSE 2.32e-03



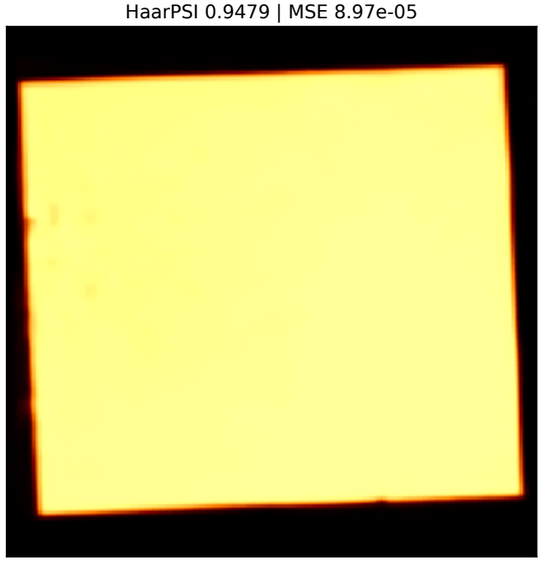
SPGL1-BPDN
PSNR 31.92 dB | SSIM 0.8249
HaarPSI 0.7322 | MSE 6.43e-04



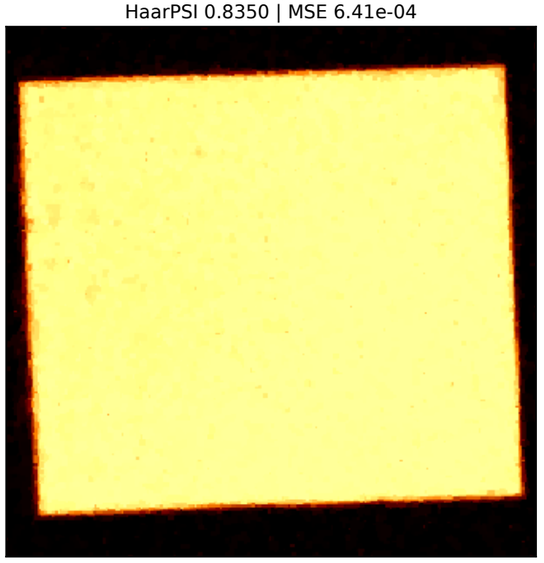
PnP-ADMM tuned-5
PSNR 39.53 dB | SSIM 0.9658
HaarPSI 0.9430 | MSE 1.11e-04



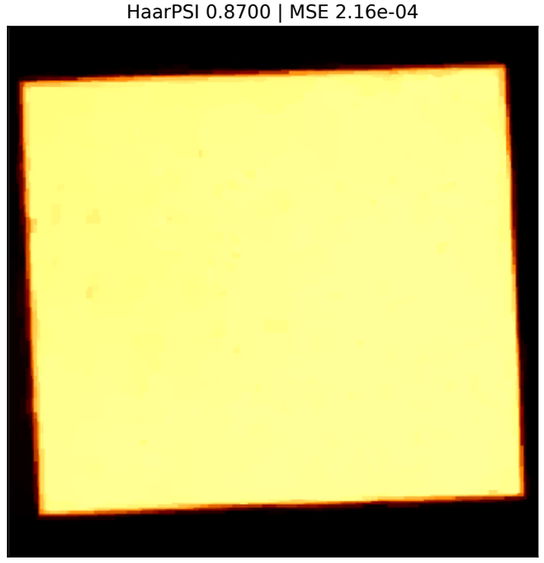
GS-PnP oracle
PSNR 40.47 dB | SSIM 0.9731
HaarPSI 0.9479 | MSE 8.97e-05



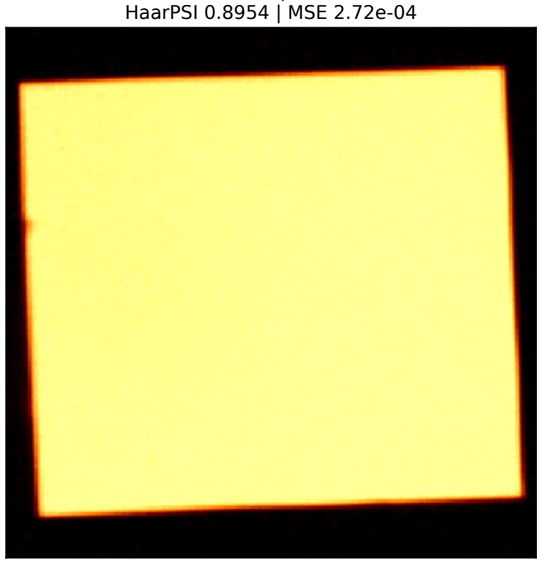
Scalar TV
PSNR 31.93 dB | SSIM 0.7828
HaarPSI 0.8350 | MSE 6.41e-04



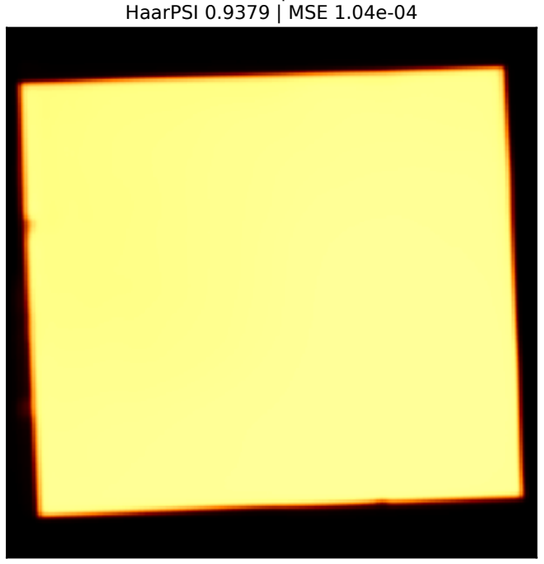
UTV-C60
PSNR 36.65 dB | SSIM 0.9440
HaarPSI 0.8700 | MSE 2.16e-04



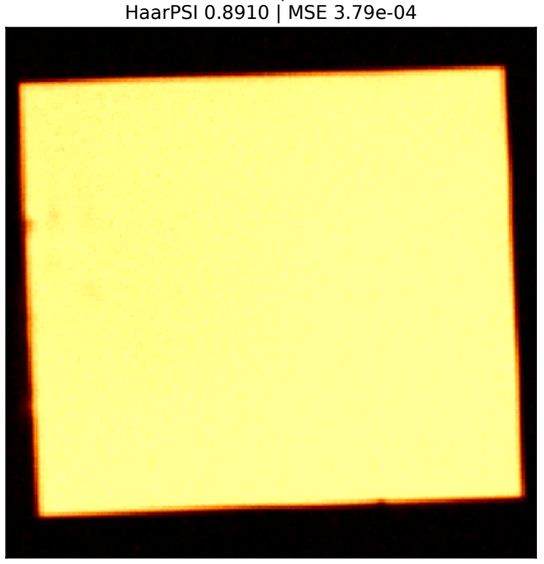
GS-PnP tuned-5
PSNR 35.66 dB | SSIM 0.8558
HaarPSI 0.8954 | MSE 2.72e-04



PnP-ADMM Tune-1 (0768)
PSNR 39.83 dB | SSIM 0.9723
HaarPSI 0.9379 | MSE 1.04e-04

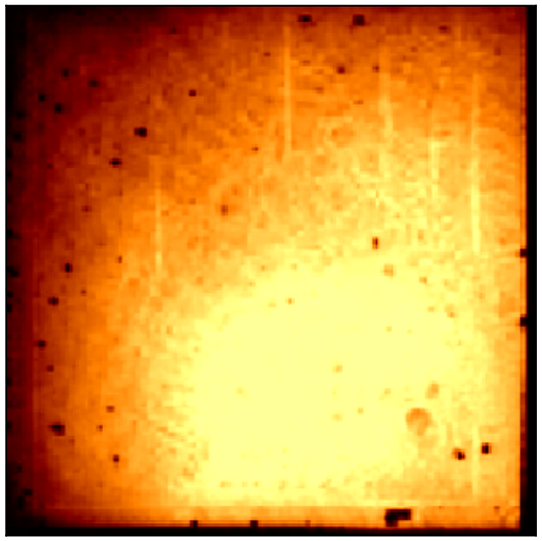


GS-PnP Tune-1 (0158)
PSNR 34.21 dB | SSIM 0.7703
HaarPSI 0.8910 | MSE 3.79e-04

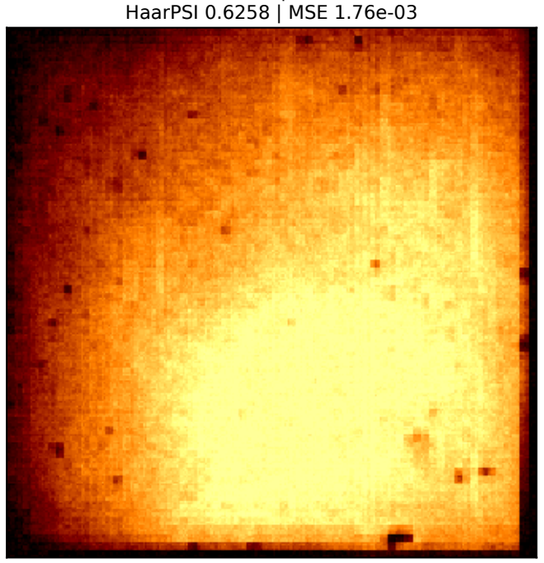


Perovskite | test image 0021 | 20% Hadamard + NPL-default noise

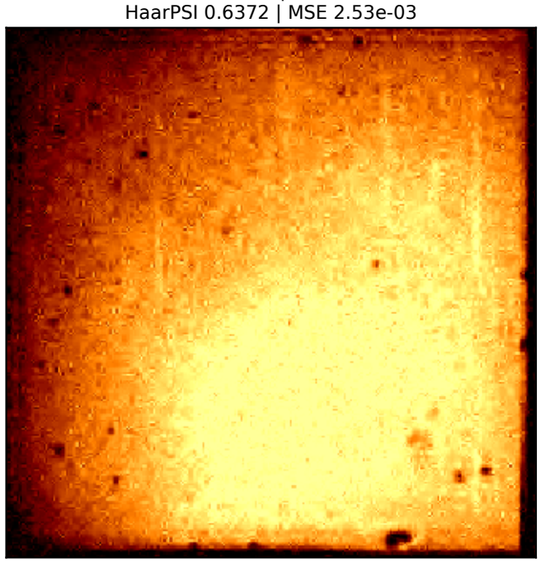
Target
Ground truth



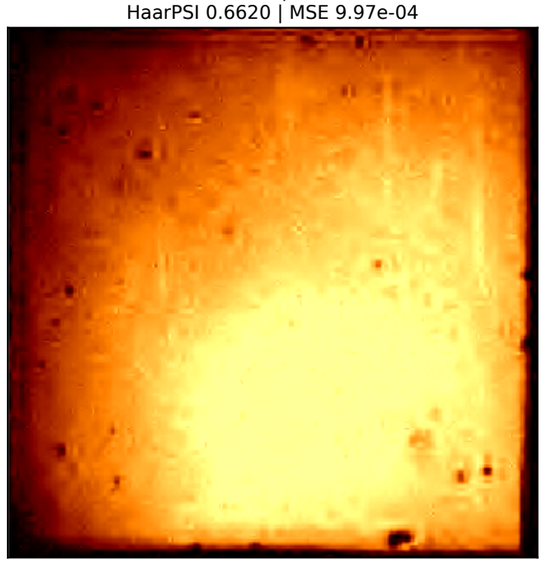
Zero-filled
PSNR 27.54 dB | SSIM 0.6188
HaarPSI 0.6258 | MSE 1.76e-03



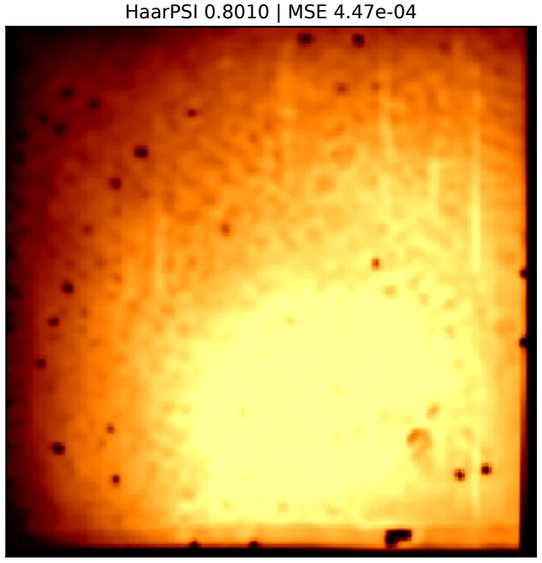
SPGL1-BP
PSNR 25.96 dB | SSIM 0.5073
HaarPSI 0.6372 | MSE 2.53e-03



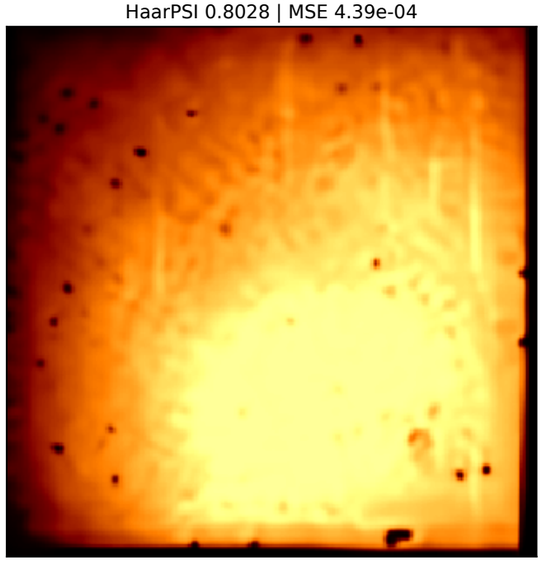
SPGL1-BPDN
PSNR 30.01 dB | SSIM 0.7672
HaarPSI 0.6620 | MSE 9.97e-04



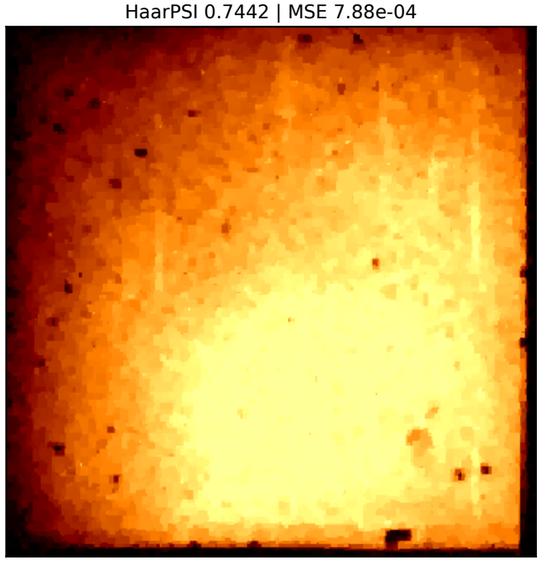
PnP-ADMM tuned-5
PSNR 33.50 dB | SSIM 0.8662
HaarPSI 0.8010 | MSE 4.47e-04



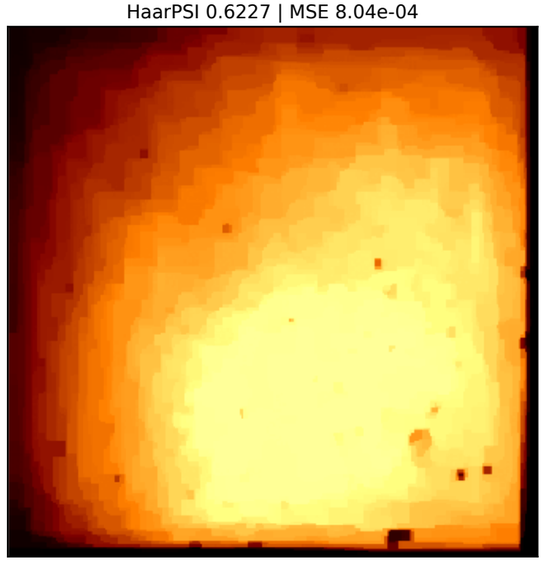
GS-PnP oracle
PSNR 33.58 dB | SSIM 0.8652
HaarPSI 0.8028 | MSE 4.39e-04



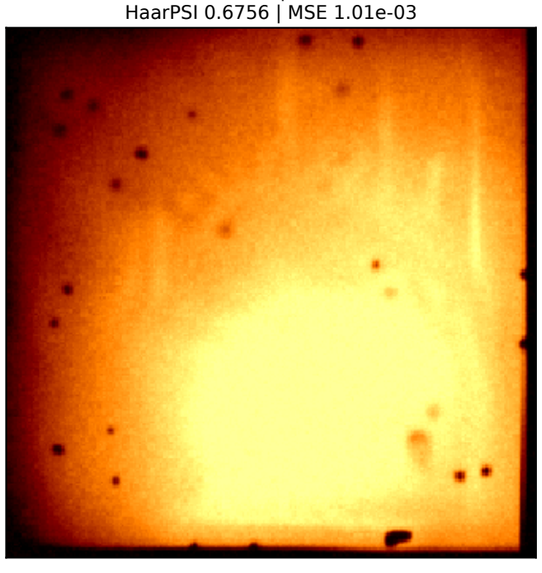
Scalar TV
PSNR 31.04 dB | SSIM 0.7853
HaarPSI 0.7442 | MSE 7.88e-04



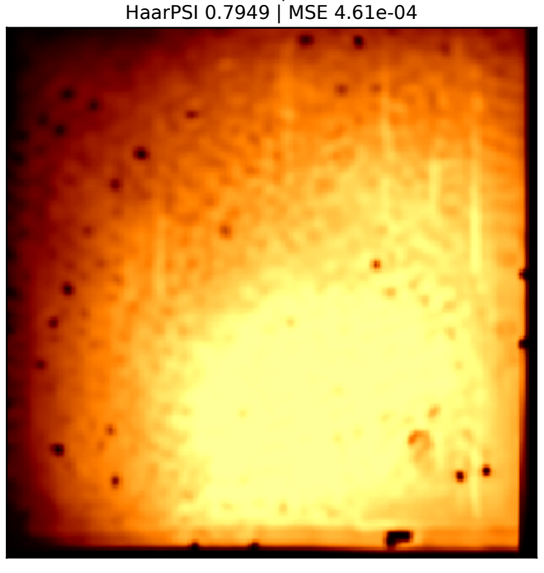
UTV-C60
PSNR 30.95 dB | SSIM 0.7955
HaarPSI 0.6227 | MSE 8.04e-04



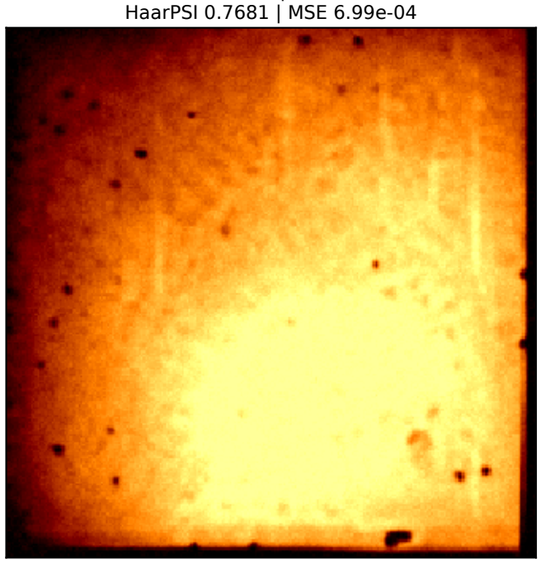
GS-PnP tuned-5
PSNR 29.97 dB | SSIM 0.6677
HaarPSI 0.6756 | MSE 1.01e-03



PnP-ADMM Tune-1 (0008)
PSNR 33.36 dB | SSIM 0.8633
HaarPSI 0.7949 | MSE 4.61e-04

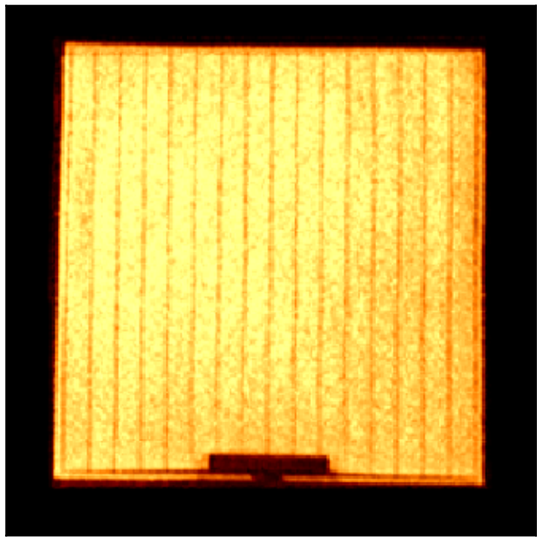


GS-PnP Tune-1 (0768)
PSNR 31.55 dB | SSIM 0.7667
HaarPSI 0.7681 | MSE 6.99e-04

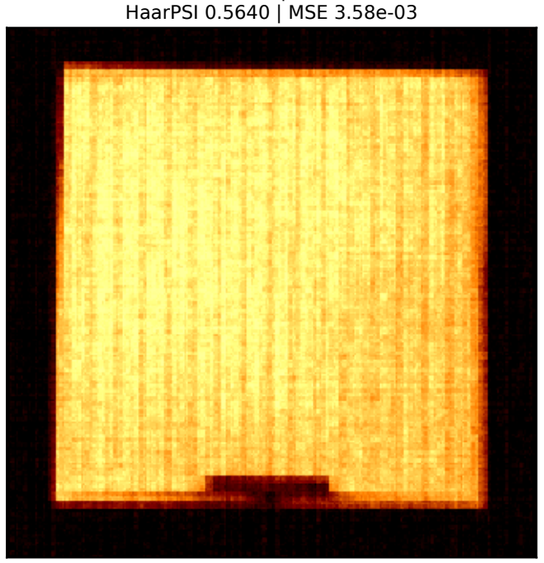


Reference | test image 0021 | 20% Hadamard + NPL-default noise

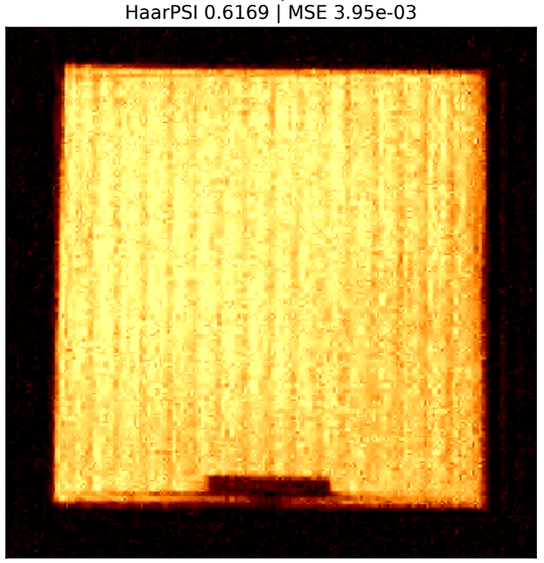
Target
Ground truth



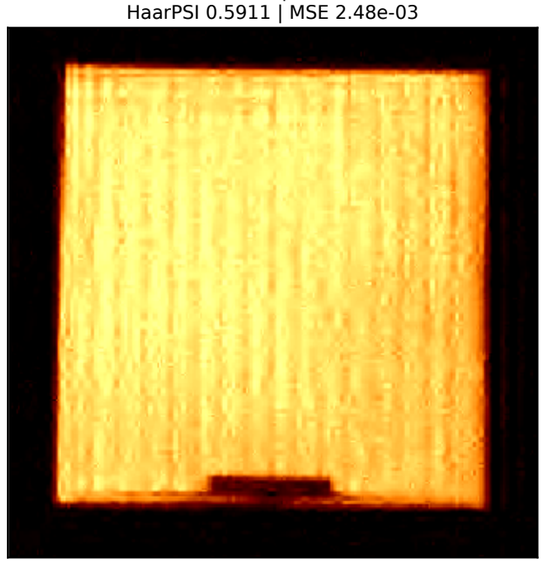
Zero-filled
PSNR 24.47 dB | SSIM 0.5090
HaarPSI 0.5640 | MSE 3.58e-03



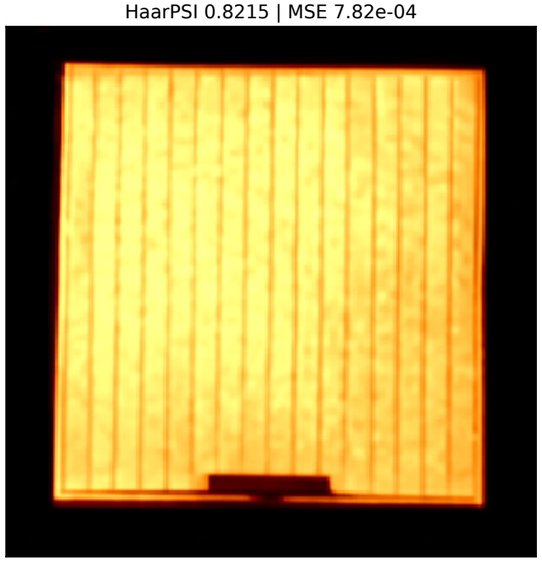
SPGL1-BP
PSNR 24.04 dB | SSIM 0.4535
HaarPSI 0.6169 | MSE 3.95e-03



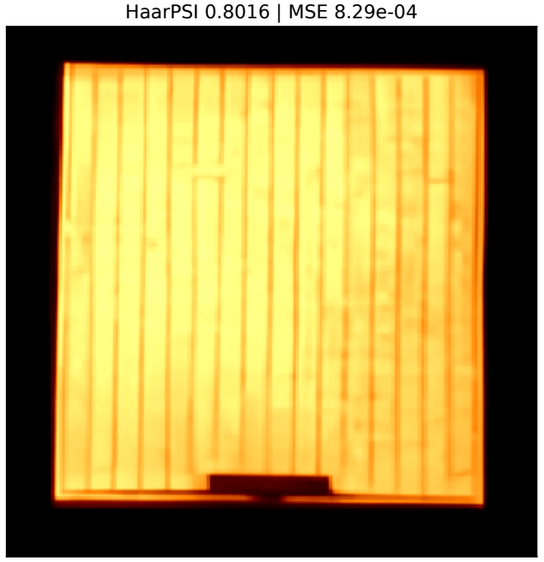
SPGL1-BPDN
PSNR 26.06 dB | SSIM 0.5797
HaarPSI 0.5911 | MSE 2.48e-03



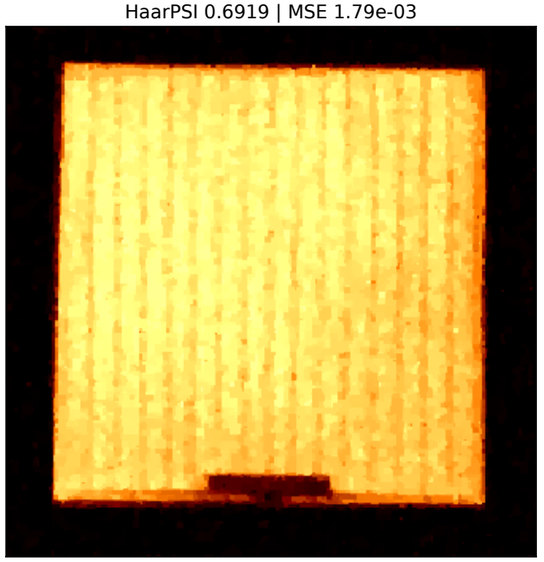
PnP-ADMM tuned-5
PSNR 31.07 dB | SSIM 0.8473
HaarPSI 0.8215 | MSE 7.82e-04



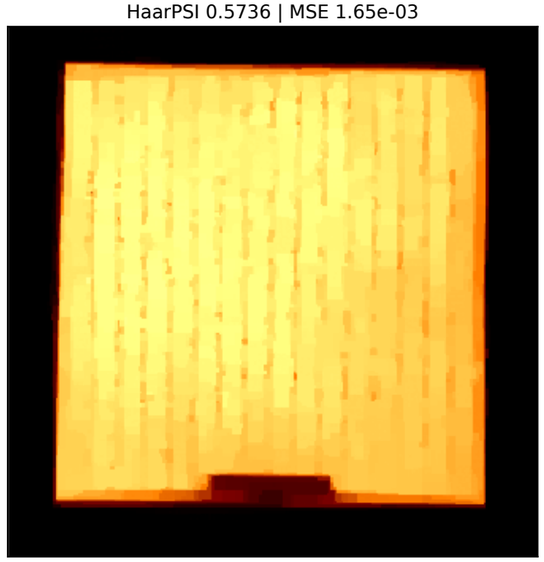
GS-PnP oracle
PSNR 30.81 dB | SSIM 0.8279
HaarPSI 0.8016 | MSE 8.29e-04



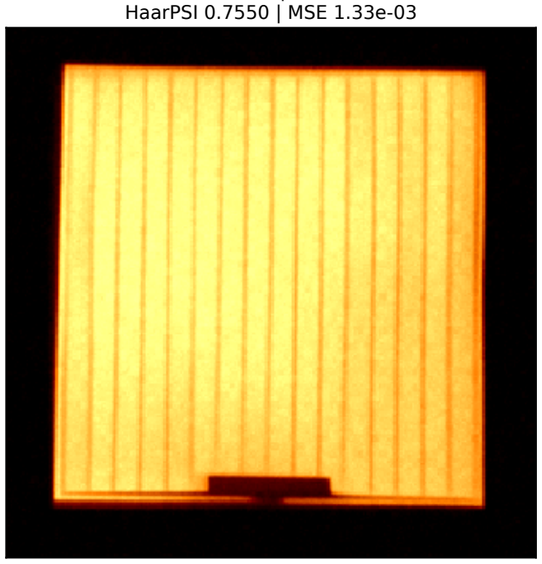
Scalar TV
PSNR 27.48 dB | SSIM 0.6419
HaarPSI 0.6919 | MSE 1.79e-03



UTV-C60
PSNR 27.83 dB | SSIM 0.6945
HaarPSI 0.5736 | MSE 1.65e-03



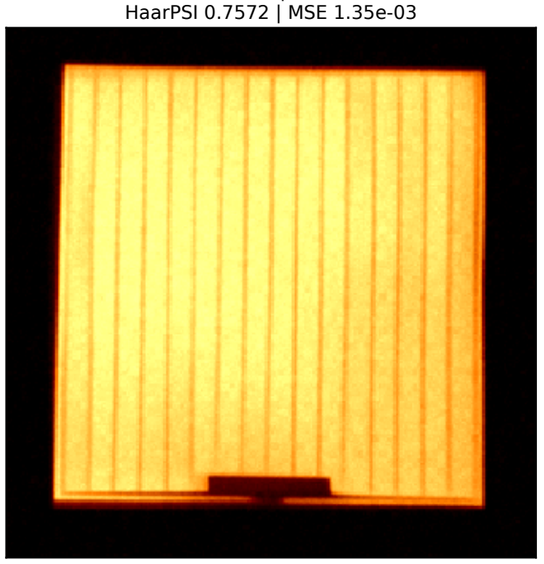
GS-PnP tuned-5
PSNR 28.76 dB | SSIM 0.7052
HaarPSI 0.7550 | MSE 1.33e-03



PnP-ADMM Tune-1 (0008)
PSNR 30.84 dB | SSIM 0.8401
HaarPSI 0.8074 | MSE 8.24e-04

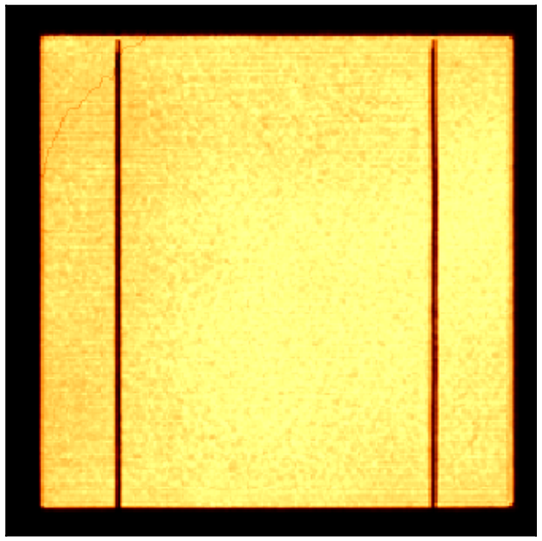


GS-PnP Tune-1 (0008)
PSNR 28.68 dB | SSIM 0.6942
HaarPSI 0.7572 | MSE 1.35e-03

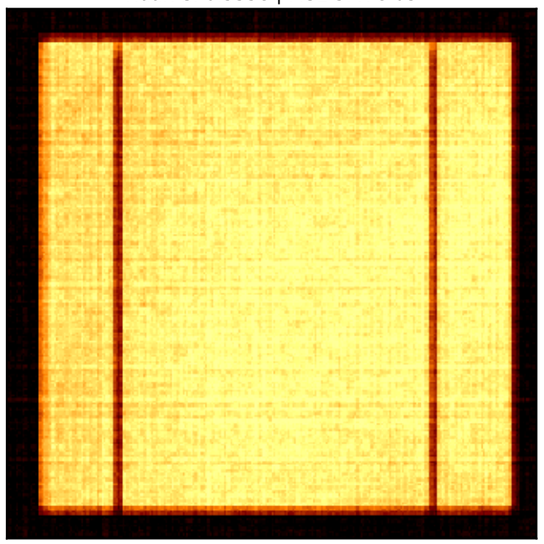


Silicone | test image 0098 | 20% Hadamard + NPL-default noise

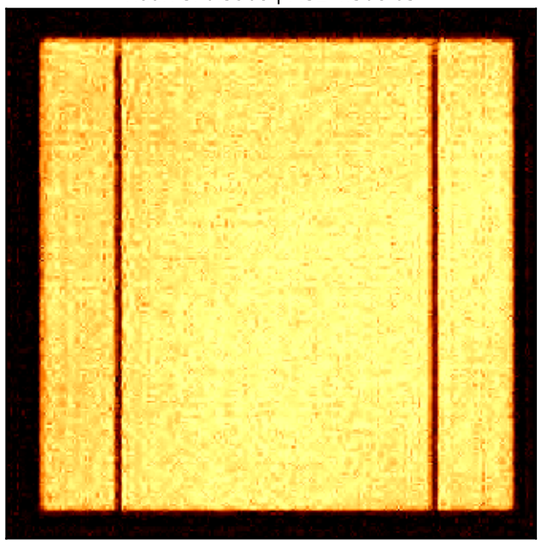
Target
Ground truth



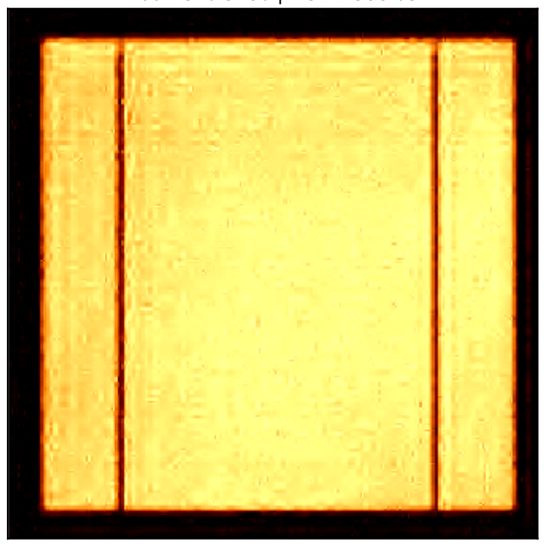
Zero-filled
PSNR 22.62 dB | SSIM 0.4408
HaarPSI 0.5358 | MSE 5.47e-03



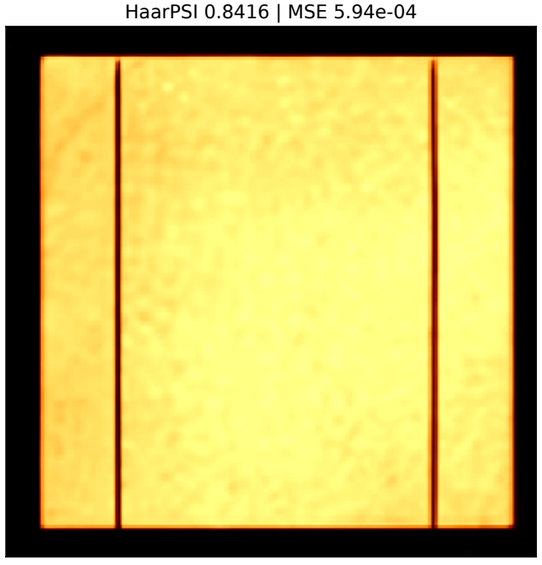
SPGL1-BP
PSNR 23.67 dB | SSIM 0.4082
HaarPSI 0.5968 | MSE 4.30e-03



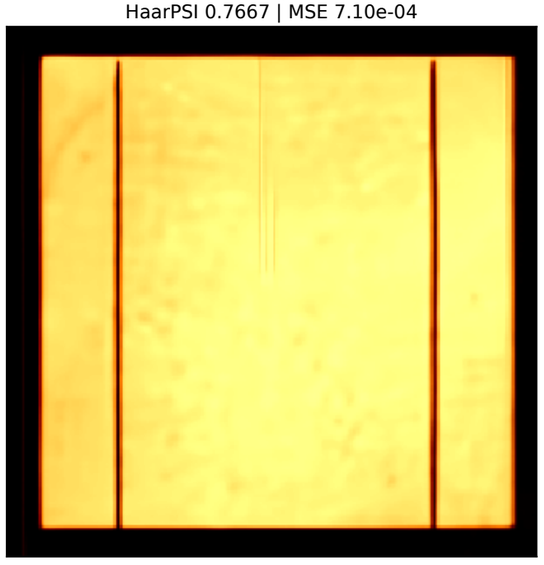
SPGL1-BPDN
PSNR 25.89 dB | SSIM 0.5728
HaarPSI 0.5780 | MSE 2.58e-03



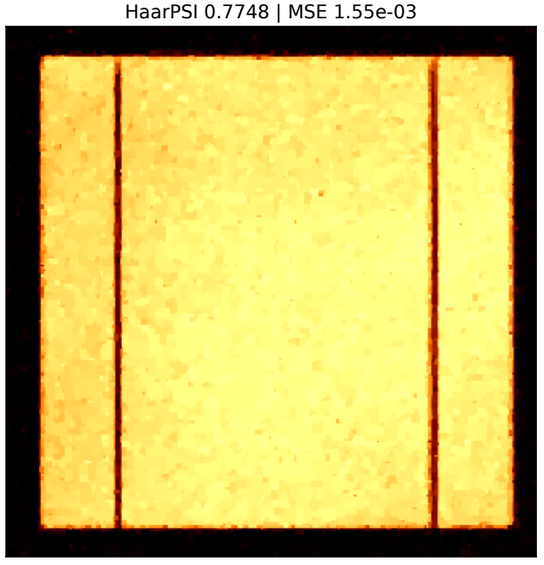
PnP-ADMM tuned-5
PSNR 32.26 dB | SSIM 0.7482
HaarPSI 0.8416 | MSE 5.94e-04



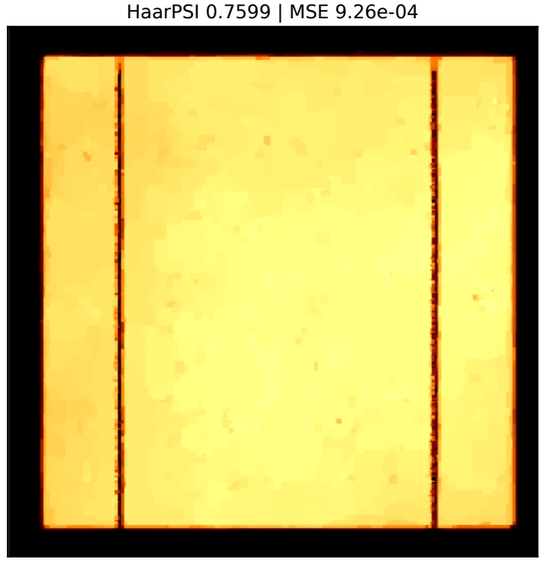
GS-PnP oracle
PSNR 31.48 dB | SSIM 0.7087
HaarPSI 0.7667 | MSE 7.10e-04



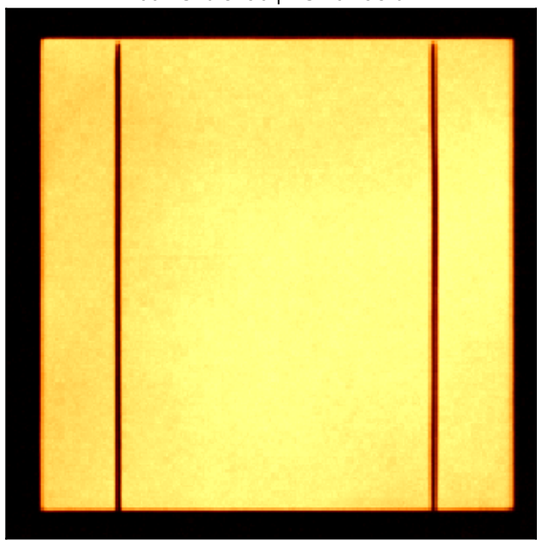
Scalar TV
PSNR 28.09 dB | SSIM 0.6475
HaarPSI 0.7748 | MSE 1.55e-03



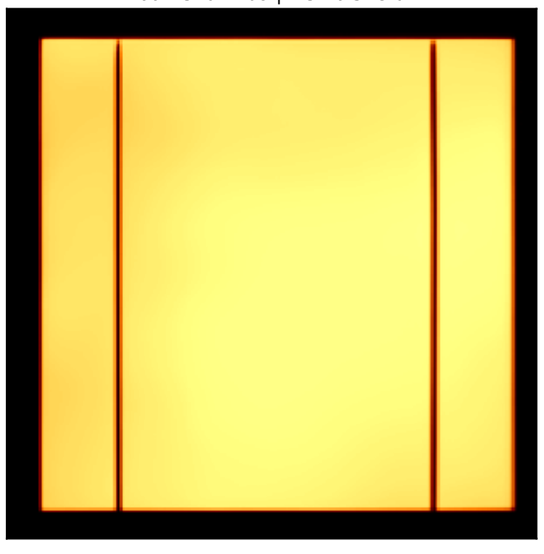
UTV-C60
PSNR 30.33 dB | SSIM 0.7334
HaarPSI 0.7599 | MSE 9.26e-04



GS-PnP tuned-5
PSNR 32.01 dB | SSIM 0.7224
HaarPSI 0.8296 | MSE 6.29e-04



PnP-ADMM Tune-1 (0008)
PSNR 31.96 dB | SSIM 0.7248
HaarPSI 0.7709 | MSE 6.37e-04



GS-PnP Tune-1 (0008)
PSNR 29.41 dB | SSIM 0.5515
HaarPSI 0.7909 | MSE 1.14e-03

